

The Singh family is building a pool. They were approved by their bank for a loan of \$85,000 with a simple interest rate of 2.25%.

1. Write an equation that models the Singh's loan.

$$A = P(1 + rt)$$

$$A = 85,000(1 + 0.0225t)$$

2. Rewrite the equation in slope-intercept form.

$$A = 85,000 + 1912.5t$$

$$A = 1912.5t + 85,000$$

3. Complete the table.

Key Feature	Value	Interpretation in Context
y-intercept	(0,85000)	When the loan begins, the balance is \$85,000.
slope	1912.5	The loan balance increases by \$1912.50 per year.

4. What is the total the Singh's will have to pay if they pay off the loan in 5 years?

$$1912.5(5) + 85,000$$

$$9,562.5 + 85,000$$

$$\$94,562.50$$

5. How many years did it take the Singh's to pay off the loan if the total they paid was \$109,000?

$$\begin{array}{rcl} 109,000 & = & 1912.5t + 85,000 \\ -85,000 & & -85,000 \end{array}$$

$$\frac{24,000}{1912.5} = \frac{1912.5t}{1912.5}$$

$$13 \approx t$$

Sophie just graduated college and is starting her first job. She is investing money into a savings account to plan for retirement. She has \$10,000 to invest in a savings account with a 4.50% APR.

6. Write an equation that models Sophie's savings account.

$$A = 10,000(1 + 0.045t)$$

7. Write the equation in slope-intercept form.

$$A = 10,000 + 450t$$

$$A = 450t + 10,000$$

8. Complete the table.

Key Feature	Value	Interpretation in Context
y-intercept	(0,10000)	There is \$10,000 in Sophie's saving account before any interest is earned.
slope	450	Each year \$450 will be added to Sophie's account.

9. How much money will be in the savings account if Sophie wants to retire 30 years later?

$$450(30) + 10,000$$

$$13,500 + 10,000$$

$$\boxed{\$23,500}$$

10. How many years would Sophie have to work to retire with \$30,000?

$$\begin{array}{rcl} 30,000 & = & 450t + 10,000 \\ -10,000 & & -10,000 \end{array}$$

$$\frac{20,000}{450} = \frac{450t}{450}$$

$$\boxed{45 \text{ years} \approx t}$$

Lakita deposited \$15,000 in a savings account with a 3.75% APR compounded monthly.

1. Write an equation that models the amount of money Lakita has in the account.

$$A = P \left(1 + \frac{r}{n} \right)^{nt} \quad A = 15,000 \left(1 + \frac{.0375}{12}^{12t} \right)$$

2. What is the balance in the account after 10 years?

$$15,000(1.003125)^{120}$$

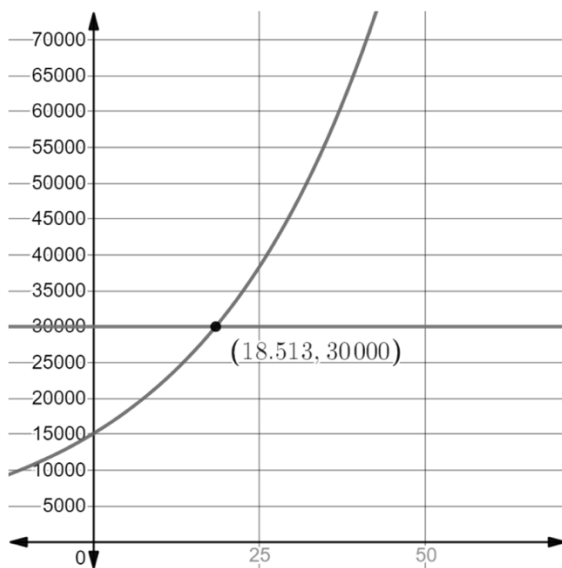
$$\$21,812.11$$

3. What is the balance in the account after 15 years?

$$15,000(1.003125)^{180}$$

$$\boxed{\$26,302.74}$$

4. Use the graph to determine when the savings account will reach \$30,000.

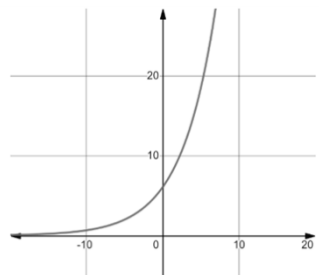


Approximately 18.5 years.

For questions 5-10, complete the table below.

	Simple Interest	Compound Interest
Investment amount and APR	\$100,000 is invested at an interest rate of 2.25% APR	\$100,000 is invested at an interest rate of 2.25% APR compounded monthly
Write the equation.	5. $A = P(1 + rt)$ $A = 100,000(1 + 0.0225t)$	6. $A = P \left(1 + \frac{r}{n} \right)^{nt}$ $A = 100,000 \left(1 + \frac{.0225}{12} \right)^{12t}$
Determine the amount after 5 years.	7. $100,000(1 + .1125)$ $\boxed{\$111,250}$	8. $100,000(1.001875)^{60}$ $\boxed{\$111,895.44}$
Determine the amount after 25 years.	9. $100,000(1 + 0.5625)$ $\boxed{\$156,250}$	10. $100,000(1.001875)^{300}$ $\boxed{\$175,413.05}$

The table shows three different functions.

Function A	Function B	Function C												
$y = 2x + 2$	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>4</td></tr><tr><td>-1</td><td>1</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>4</td></tr></table>	x	y	-2	4	-1	1	0	0	1	1	2	4	
x	y													
-2	4													
-1	1													
0	0													
1	1													
2	4													

1. Determine which type of function that each represents.

Function A	Function B	Function C
☐ Exponential ☒ Linear ☐ Quadratic	☐ Exponential ☐ Linear ☒ Quadratic	☒ Exponential ☐ Linear ☐ Quadratic

2. Which function has the largest y -intercept?

Function C

3. Compare the ranges of the functions and indicate any similarities and differences.

The range of Function A is all real numbers. Functions B and C have similar ranges, but not the same. Function B's range is $y \geq 0$, and Function C's range is $y > 0$.

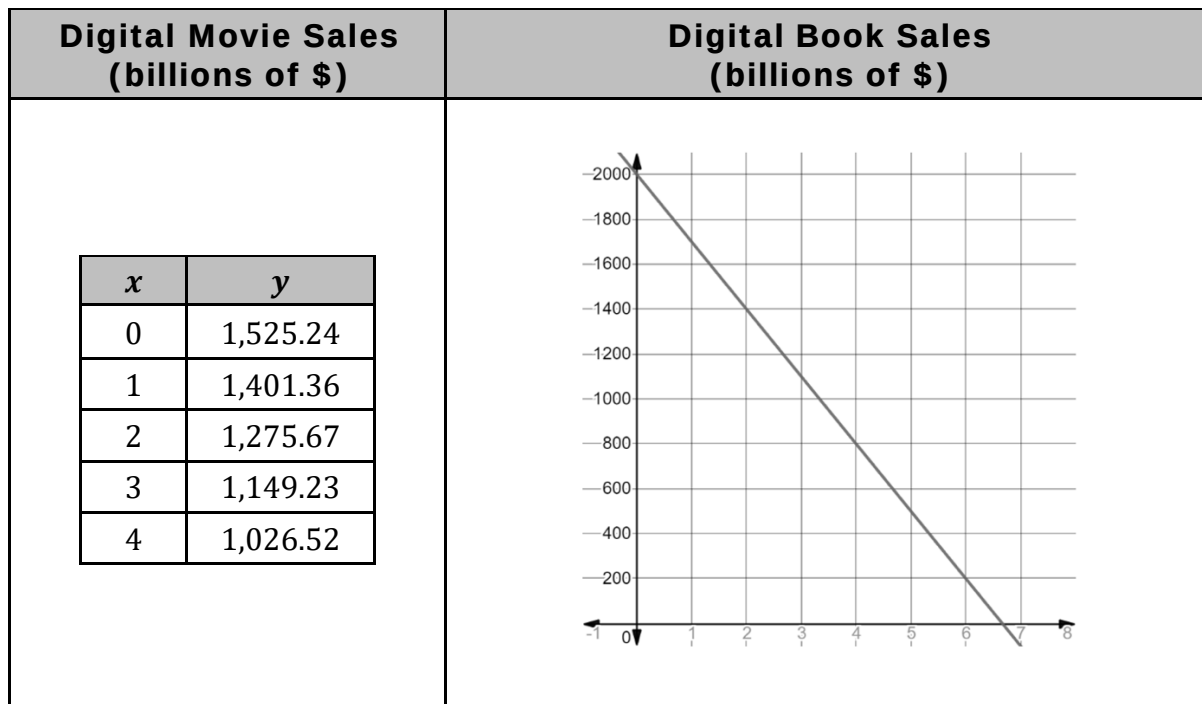
4. Compare the domains of the functions and indicate any similarities and differences.

All three functions have the same domain – all real numbers.

5. As $x \rightarrow \infty$, which function has larger values of y ?

Function C

The table shows a table of values representing the revenue, in billions of dollars, from sales of digital movies, where $x = 0$ is 2015, and a graph representing the revenue, in billions of dollars, from sales of digital books, where $x = 0$ is 2015.

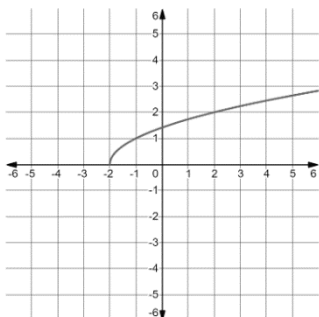


6. Approximate the x -intercept for the function that represents the sale of digital books.
(6.6,0)
7. Interpret the x -intercept of the function that represents the sale of digital books.
Approximately 6.6 years after 2015, there will be no revenue from sales of digital books.
8. What is the domain of the function representing the sale of digital books?
 $\{x|x \in \mathbb{R}\}$
9. Which function has a larger y -intercept?
Digital book sales
10. What type of function best describes the table of values that represents digital movie sales? Explain your choice.

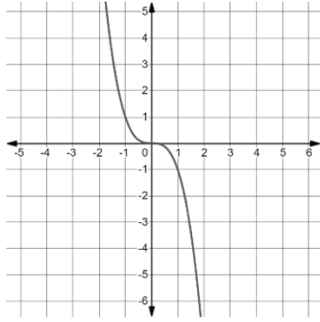
- Exponential
 - Linear
 - Quadratic

The rate of change is fairly consistent from year to year.

Determine which family of functions each transformation is part of.

Representation of Transformation		Equation of Parent Function													
1.	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>16</td></tr><tr><td>-1</td><td>4</td></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>4</td></tr><tr><td>2</td><td>16</td></tr></table>	x	y	-2	16	-1	4	0	0	1	4	2	16	☐ $y = x$ ☒ $y = x^2$ ☐ $y = x^3$ ☐ $y = \sqrt{x}$	☐ $y = x$ ☐ $y = 2^x$ ☐ $y = \frac{1}{2}^x$
x	y														
-2	16														
-1	4														
0	0														
1	4														
2	16														
2.		☐ $y = x$ ☐ $y = x^2$ ☐ $y = x^3$ ☒ $y = \sqrt{x}$	☐ $y = x$ ☐ $y = 2^x$ ☐ $y = \frac{1}{2}^x$												
3.	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>5</td></tr><tr><td>-1</td><td>7</td></tr><tr><td>0</td><td>9</td></tr><tr><td>1</td><td>11</td></tr><tr><td>2</td><td>13</td></tr></table>	x	y	-2	5	-1	7	0	9	1	11	2	13	☒ $y = x$ ☐ $y = x^2$ ☐ $y = x^3$ ☐ $y = \sqrt{x}$	☐ $y = x$ ☐ $y = 2^x$ ☐ $y = \frac{1}{2}^x$
x	y														
-2	5														
-1	7														
0	9														
1	11														
2	13														
4.	$y = -3x^3$	☐ $y = x$ ☐ $y = x^2$ ☒ $y = x^3$ ☐ $y = \sqrt{x}$	☐ $y = x$ ☐ $y = 2^x$ ☐ $y = \frac{1}{2}^x$												
5.	<table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>9</td></tr><tr><td>-1</td><td>7</td></tr><tr><td>0</td><td>5</td></tr><tr><td>1</td><td>7</td></tr><tr><td>2</td><td>9</td></tr></table>	x	y	-2	9	-1	7	0	5	1	7	2	9	☐ $y = x$ ☐ $y = x^2$ ☐ $y = x^3$ ☐ $y = \sqrt{x}$	☒ $y = x$ ☐ $y = 2^x$ ☐ $y = \frac{1}{2}^x$
x	y														
-2	9														
-1	7														
0	5														
1	7														
2	9														

6.



- ☐ $y = x$
☐ $y = x^2$
☒ $y = x^3$
☐ $y = \sqrt{x}$

- ☐ $y = |x|$
☐ $y = 2^x$
☐ $y = \frac{1}{2}^x$

7.

x	-2	-1	0	1	2
y	-4	-2	-1	-0.5	-0.25

- ☐ $y = x$
☐ $y = x^2$
☐ $y = x^3$
☐ $y = \sqrt{x}$

- ☐ $y = |x|$
☐ $y = 2^x$
☒ $y = \frac{1}{2}^x$

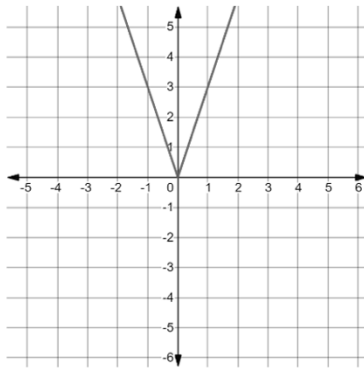
8.

x	-2	-1	0	1	2
y	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	1	2

- ☐ $y = x$
☐ $y = x^2$
☐ $y = x^3$
☐ $y = \sqrt{x}$

- ☐ $y = |x|$
☒ $y = 2^x$
☐ $y = \frac{1}{2}^x$

9.



- ☐ $y = x$
☐ $y = x^2$
☐ $y = x^3$
☐ $y = \sqrt{x}$

- ☒ $y = |x|$
☐ $y = 2^x$
☐ $y = \frac{1}{2}^x$

10.

$$y = -\frac{3}{4}x + 4$$

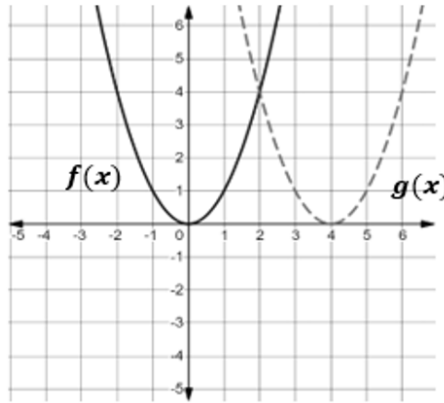
- ☒ $y = x$
☐ $y = x^2$
☐ $y = x^3$
☐ $y = \sqrt{x}$

- ☐ $y = |x|$
☐ $y = 2^x$
☐ $y = \frac{1}{2}^x$

Algebra 1
Unit 13
Lesson 5 Homework

Name Answer Key
Date _____
Period _____

The graphs of $y = f(x)$ and $y = g(x)$ are shown.



1. Describe how to shift the graph of $f(x)$ so that it coincides with the graph of $g(x)$.

Moves 4 units to the right

2. Use the graph to complete the table of values for $f(x)$ and $g(x)$.

$f(x)$		$g(x)$	
x	y	x	y
-1	1	3	1
0	0	4	0
1	1	5	1

3. Explain the change in the x -values from $f(x)$ to $g(x)$.

Add 4 to each x from $f(x)$ to $g(x)$.

4. Complete the statement:

The graph of $g(x)$ is a ☒ horizontal
☐ vertical shift of $f(x)$.

5. Complete the statement:

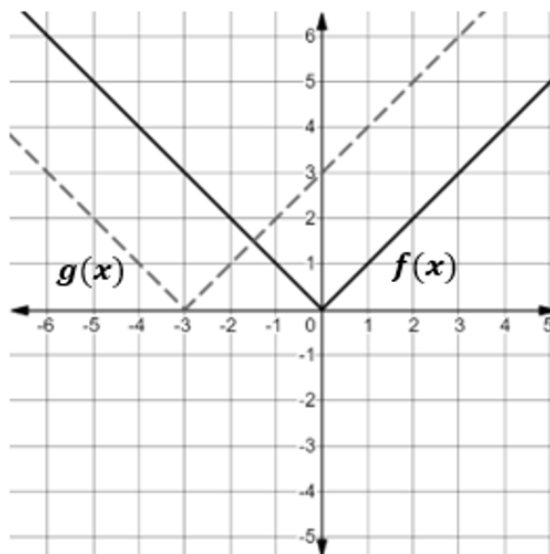
The shift is seen in the table by ☒ adding
☐ subtracting 4 to each ☒ x -value
☐ y -value

for each corresponding ☐ x -value.
☒ y -value.

Complete the table for questions 6-8. The first row has been completed.

Original Equation	Original Function	Shift	Transformed Function
$y = x^2$	$f(x) = x^2$	Left 2	$f(x + 2) = (x + 2)^2$
6. $y = x $	$f(x) = x $	Right 4	$f(x - 4) = x - 4 $
7. $y = x^3$	$f(x) = x^3$	Left 5	$f(x + 5) = (x + 5)^3$
8. $y = \sqrt{x}$	$f(x) = \sqrt{x}$	Right 3	$f(x - 3) = \sqrt{x - 3}$

The graphs of $y = f(x)$ and $y = g(x)$ are shown.



9. How does the graph of $f(x)$ compare to the graph of $g(x)$?

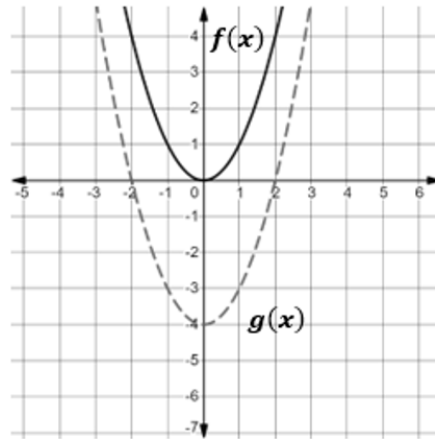
The graph of $f(x)$ is 3 units to the right of $g(x)$.

10. Two function descriptions are shown. Which function description matches the transformation on the graph? Justify your choice.

- ☐ $g(x) = f(x - 3)$
- ☒ $g(x) = f(x + 3)$

The vertex of $f(x)$ moved three units to the left, resulting in $g(x)$.

The graphs of $y = f(x)$ and $y = g(x)$ are shown.



1. Describe how to shift the graph of $f(x)$ so that it coincides with the graph of $g(x)$.

Moves down 4 units

2. Use the graph to complete the table of values for $f(x)$ and $g(x)$.

$f(x)$		$g(x)$	
x	y	x	y
-1	1	-1	-3
0	0	0	-4
1	1	1	-3

3. Explain the change in the y -values from $f(x)$ to $g(x)$.

Subtract 4 from the y -values.

4. Complete the statement:

The graph of $g(x)$ is a ☐ horizontal ☒ vertical shift of $f(x)$.

5. Complete the statement:

The shift is seen in the table by ☐ adding ☒ subtracting 4 to each ☐ x -value ☒ y -value

for each corresponding ☐ x -value. ☒ y -value.

For 6-7, complete the statements to describe how the graph of $g(x) = f(x) + 7$ compares to the graph of $f(x) = |x - 1|$.

6. The graph of $g(x)$ is a

☐ vertical

☐ horizontal

 shift of $f(x)$.

7. The graph of $g(x)$ is shifted 7 units

☐ up from

☐ down from
☐ to the right of
☐ to the left of

 the graph of $f(x)$.

8. Describe the effect on the x -values of $B(x) = |x + 5|$ for the transformation $C(x) = B(x) - 7$.

$C(x)$ is 7 below $B(x)$, the x -values are the same.

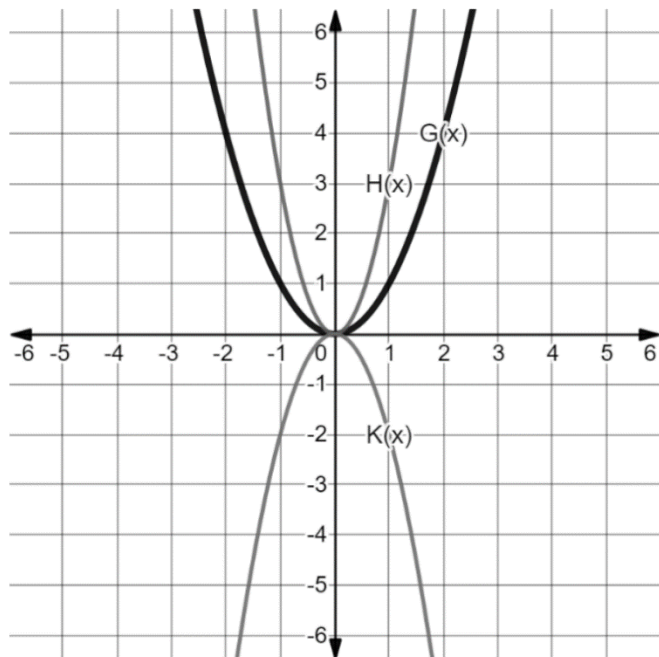
9. Describe the effect on the y -values of $p(x) = \frac{4}{5}x + 2$ for the transformation $r(x) = p(x) + 8$.

$r(x)$ is 8 above $p(x)$, the y -values are all 8 more than the y -values of $p(x)$.

10. Describe how the graph of $M(x) = N(x + 10)$ compares to the graph of $N(x) = x^2 + 3$.

$m(x)$ is 10 units to the left of $n(x)$.

The functions $G(x)$, $H(x)$, and $K(x)$ are shown on the graph, where $G(x) = x^2$, $H(x) = 3x^2$, and $K(x) = -2x^2$. The table of values for each function are also shown.



$y = G(x)$		$y = H(x)$		$y = K(x)$	
x	y	x	y	x	y
-3	9	-3	27	-3	-18
-2	4	-2	12	-2	-8
-1	1	-1	3	-1	-2
0	0	0	0	0	0
1	1	1	3	1	-2
2	4	2	12	2	-8
3	9	3	27	3	-18

Complete the statements below.

- The coefficient of $G(x) = x^2$ is **1**.
- The coefficient of $H(x) = 3x^2$ is **3**.
- The y -values of $H(x)$ can be found by multiplying the y -values of $G(x)$ by **3**.
- The coefficient of $K(x) = -2x^2$ is **-2**.
- The y -values of $K(x)$ can be found by multiplying the y -values of $G(x)$ by **-2**.

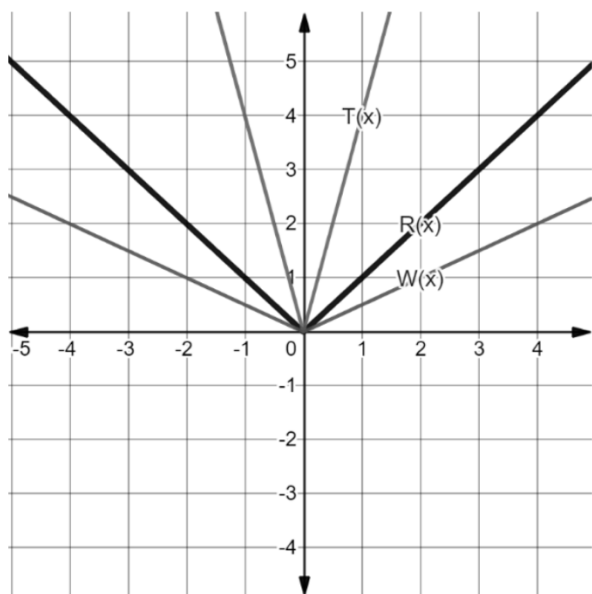
The table of values of the function $y = A(x)$ is shown.

x	-6	-4	-2	0	2	4	6
y	4	7	10	13	16	19	22

Complete the sentence to describe the effect on the x -values for the function $y = A(3x)$.

6. The x -values for $A(x)$ will be
☐ multiplied
☒ divided
 by 3.

The functions $R(x)$, $T(x)$, and $W(x)$ are shown on the graph, where $R(x) = |x|$, $T(x) = |4x|$, and $W(x) = |0.5x|$. The table of values for each function are also shown.



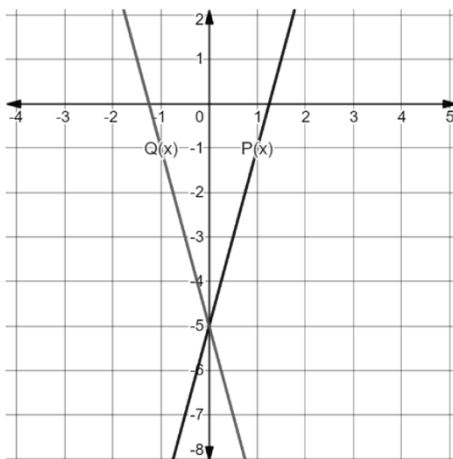
$y = R(x)$		$y = T(x)$		$y = W(x)$	
x	y	x	y	x	y
-2	2	$-\frac{1}{2}$	2	-4	2
-1	1	$-\frac{1}{4}$	1	-2	1
0	0	0	0	0	0
1	1	$\frac{1}{4}$	1	2	1
2	2	$\frac{1}{2}$	2	4	2

Complete the statements below.

7. The x -values of $T(x)$ can be found by ☐ multiplying ☒ dividing the x -values of $R(x)$ by 4.
8. The x -values of $W(x)$ can be found by ☐ multiplying ☒ dividing the x -values of $R(x)$ by $\frac{1}{2}$ or multiply by 2.

The functions $P(x)$ and $Q(x)$ are shown on the graph where $Q(x) = P(-x)$. The table of values for each function are also shown.

$y = P(x)$		$y = Q(x)$	
x	y	x	y
-2	-13	-2	3
-1	-9	-1	-1
0	-5	0	-5
1	-1	1	-9
2	3	2	-13



9. What is the effect on the table of values when the x in $P(x)$ was replaced with $-x$ to create $Q(x)$?
Explanations may vary. The y -values changed to the opposite x -value.
10. What is the effect on the graph when the x in $P(x)$ was replaced with $-x$ to create $Q(x)$?
reflects over the y -axis

Suni is biking 20 miles. To fuel her bike ride, she will need to consume 738 calories. She plans to bring sports drink and granola bars. Sports drink bottles are 158 calories and granola bars are 132 calories.

1. $158s + 132g$ is a(n)

- ☐ function.
- ☐ equation.
- ★ ☒ expression.

2. The values 158 and 132 are

- ★ ☒ coefficients.
- ☐ terms.
- ☐ constants.

3. The monomials $158s$ and $132g$ are

- ☐ coefficients.
- ★ ☒ terms.
- ☐ constants.

4. $158s + 132g = 738$ is a(n)

- ☐ function.
- ★ ☒ equation.
- ☐ expression.

5. Determine the amount of granola bars Suni will need if she has three bottles of sports drink.

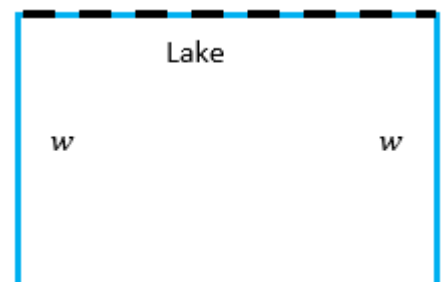
$$158(3) + 132g = 738$$

$$\begin{array}{r} 474 + 132g = 738 \\ -474 \quad \quad -474 \end{array}$$

$$\frac{132g}{132} = \frac{264}{132}$$

2 granola bars

A rancher has 3000 feet of fencing to enclose his property, which borders a lake. The diagram shows the width, w , of the property and the lake that his property borders. The fencing is not used along the lake.



Explain what each quantity represents in this context.

6. $3000 - 2w$ = amount of fence left for last side.

3000 = total fence to be used

$2w$ = two sides that will use fence, where w is the length of those sides.

7. $w(3000 - 2w)$

Width times amount of fence used for the last side.

Luis started a retirement savings fund when he began his first job in 2015. The amount of money in the retirement savings fund since 2015 can be represented by the equation $y = 15,000(1 + 0.025)^x$, where y is the amount in the retirement savings fund and x is the number of years since 2015.

8. The equation $y = 15,000(1 + 0.025)^x$ represents

- ☒ exponential growth.
☐ exponential decay.

9. The rate of

- ☐ decay
☒ growth

is 2.5 %.

10. What does the coefficient 15,000 represent?

Initial amount in savings account.

To make a 15% acid solution, a chemist mixes 3 parts of 4% acid solution with a 20% acid solution. The expression $3(0.04x) + 0.20y$ represents the mixture, where x is the number of ounces of 4% solution and y is the number of ounces of the 20% solution.

Explain what each quantity means.

1. $(0.04x)$

Amount of acid solution per part of 4% solution.

2. $3(0.04x)$

Amount of acid solution mixed into mixture of 4% solution.

3. $0.20y$

Amount of acid solution per part of 20% solution.

Since 2015, the number of independently contracted food delivery drivers can be modeled by the equation $y = 1,250,000(1.073)^x$.

4. What does the x variable represent?

Number of years since 2015.

5. What does the y variable represent?

Total amount of contracted drivers an " x " number of years after 2015.

6. What does the quantity 1,250,000 represent?

Initial number of drivers in 2015.

7. What is the constant percent rate of change?

107.3%

The equation $y = -2(x - 4)^2 + 45$ models the height, y , in feet, of a ball, x seconds after it is thrown into the air.

Explain what each quantity represents.

8. $x - 4$

the number of seconds away the ball's height is from the maximum

9. 45

maximum height of the ball

Jovi is buying cereal for \$3.25 per box. He has a coupon for two free cereals. He pays \$4.75 for a gallon of milk. The expression $3.25(x - 2) + 4.75$ can be used to represent the overall cost.

10. What does the quantity $3.25(x - 2)$ represent?

Cost of cereal boxes purchased.

Given a written description, identify the function as linear, quadratic, or exponential. Explain your choice.

1. An investment grows at a constant percent rate of 2.25% APR.

Exponential: constant percent will always be exponential.

2. The Occupational Safety and Health Administration (OSHA) allows for stairs to have a rise of 9 inches and a run of 10 inches.

Linear: constant ratio of 9:10.

3. A ball is thrown from a height of 5 feet. Four seconds later, it reaches a maximum height of 25 feet.

Quadratic: follows a quadratic curve it will go up then fall back down with change in direction.

4. A rare species of fish experiences a population decline at a constant percent rate of 4.5%.

Exponential: constant percent will always be exponential.

Given a table of values, identify the function as linear, quadratic, or exponential. Explain your choice.

5.

x	y
0	120
1	111
2	102
3	93
4	84

Linear, constant 1st difference

6.

x	y
0	165
1	179
2	195
3	211
4	230

Exponential, no constant 1st or 2nd difference but ratio ≈ 0.92 .

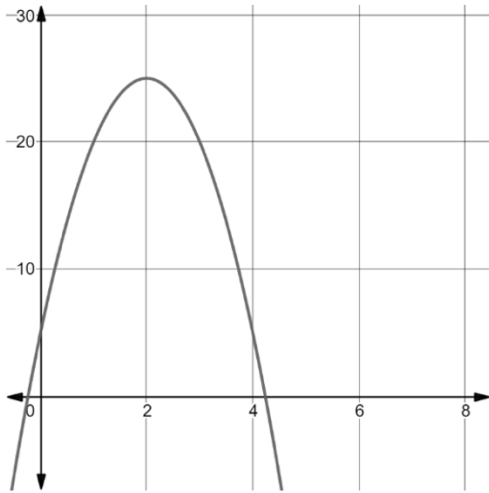
7.

x	y
1	20
2	35
3	40
4	35
5	20

Quadratic, constant 2nd difference

Given a graph, identify the function as linear, quadratic, or exponential. Explain your choice.

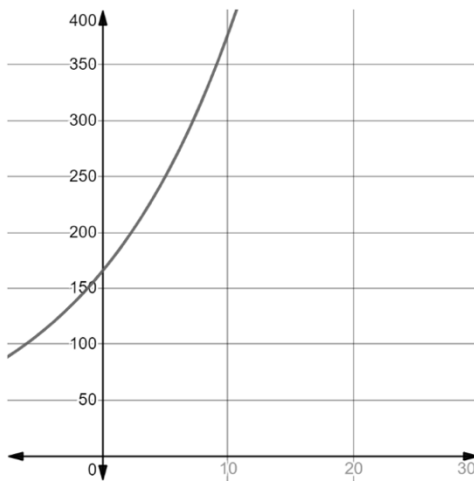
8.



Quadratic

- follows quadratic curve
- constant second difference in y -values

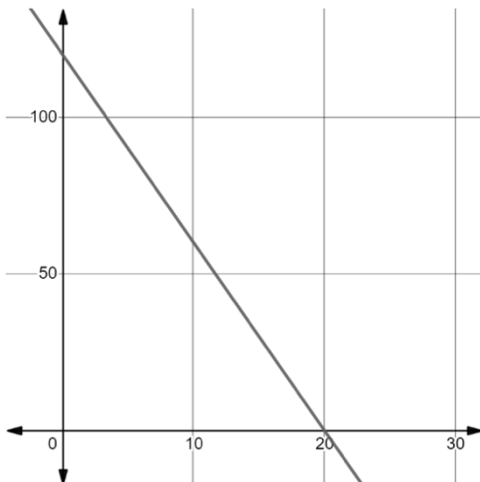
9.



Exponential

- constant ratio from one y -value to the next
- follows exponential curve

10.



Linear

- straight line
- constant slope

The table shows the number of tons of blueberries that were imported to the US from 2005 to 2013.

Year	Blueberries Imported (tons)
2005	26,319
2006	32,602
2007	35,125
2008	52,111
2009	60,450
2010	76,743
2011	87,336
2012	96,367
2013	103,502

1. Determine the average rate of change from 2005 to 2013.

$$\begin{array}{l} (2005, 26,319) \\ (2013, 103,502) \end{array} \quad \frac{103,502 - 26,319}{2013 - 2005} = \frac{77,183}{8} = 9,647.875$$

2. Interpret the average rate of change for the interval 2005 to 2013 in terms of the context.

From 2005 to 2013, an average of 9,647.875 more tons of blueberries were imported each year.

3. Determine the average rate of change from 2007 to 2010.

$$\begin{array}{l} (2007, 35125) \\ (2010, 76743) \end{array} \quad \frac{76,743 - 35,125}{2010 - 2007} = \frac{41,618}{3} = 13,872.\bar{6}$$

4. Interpret the average rate of change for the interval 2007 to 2010 in terms of the context.

From 2007 to 2010, an average of 13,872. $\bar{6}$ more tons of blueberries were imported each year.

5. How does the average rate of change for the interval 2007 to 2010 compare to the average rate of change for the interval 2005 to 2013?

The average for 2007 – 2010 was larger by 4,224.792 tons.

The function $M(x) = 5x^2 - 160x + 6179$ models the number of miles, in millions, U.S. passengers traveled on a train for the period of 1970 to 2000, where x is the number of years since 1970.

6. What is the average rate of change, in miles traveled by U.S. passengers per year, for the period 1970 to 1975?

$$1970 \rightarrow x = 0$$

$$m(0) = 6,179$$

$$1975 \rightarrow x = 5$$

$$m(5) = 5(25) - 160(5) + 6,179$$

$$m(5) = 5504$$

$$\frac{5504-6179}{5-0} = \frac{-675}{5} = -135$$

7. Interpret the average rate of change for the interval 1970 to 1975 in terms of the context.

From 1970 to 1975, the average decreased by 135 million miles yearly.

8. What is the average rate of change, in miles traveled by U.S. passengers per year, for the period 1975 to 1980?

$$m(5) = 5504$$

$$1975 \rightarrow x = 5$$

$$m(10) = 500 - 1600 + 6179$$

$$1980 \rightarrow x = 10$$

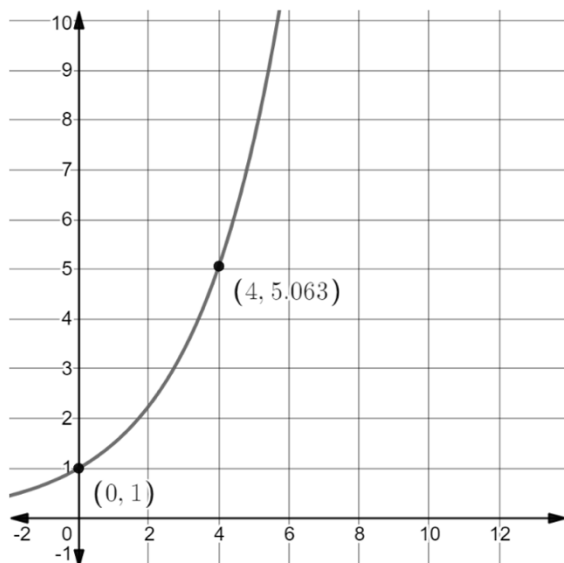
$$m(10) = 5079$$

$$\frac{5079-5504}{10-5} = \frac{-425}{5} = -85$$

9. Interpret the average rate of change for the interval 1975 to 1980 in terms of the context.

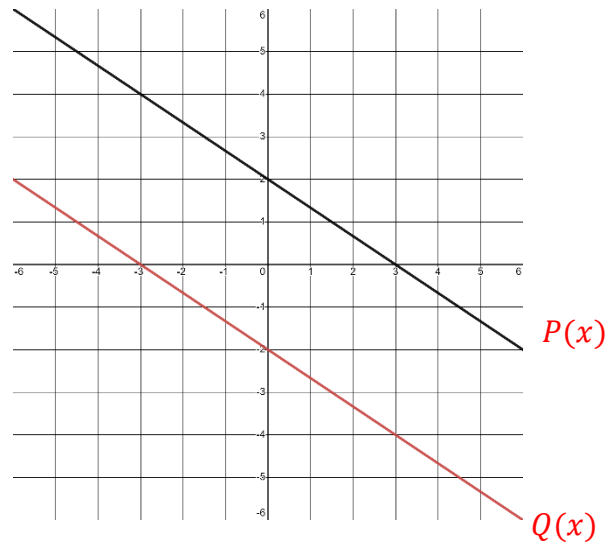
From 1975 to 1980, the average decreased by 85 million miles yearly.

10. The graph of a function is shown. Points $(0, 1)$ and $(4, 5.063)$ are on the function. Calculate the average rate of change for the line that passes through the points $(0, 1)$ and $(4, 5.063)$.



$$\frac{5.063-1}{4-0} = \frac{4.063}{4} = 1.01575$$

The graph of the function $P(x)$ is shown.



1. Write the function, $P(x)$, that represents the line.

$$P(x) = -\frac{2}{3}x + 2$$

2. Draw a line that is parallel to $P(x)$ with a y -intercept of -2 . Label this line $Q(x)$.
3. Describe how $P(x)$ can be shifted to coincide with $Q(x)$.

Shift line down 4.

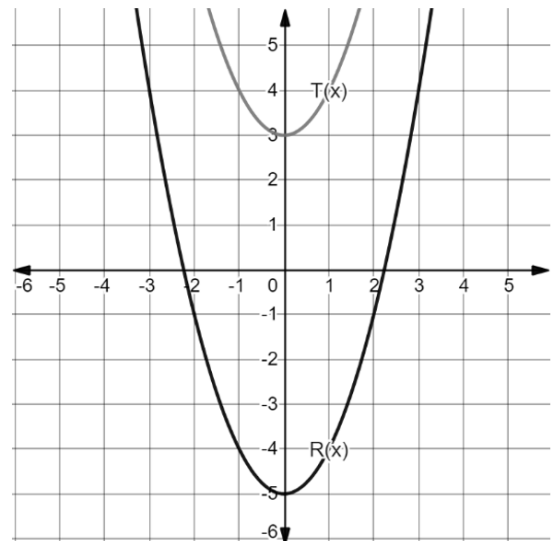
4. Write the function $Q(x)$ in terms of $P(x)$.

$$Q(x) = P(x) - 4$$

5. The graphs of $R(x)$ and $T(x)$ are shown on the graph.

Write the transformation by completing the equation.

$$T(x) = R(x) + 8$$



The functions $h(x)$ and $j(x)$ are shown on the graph.

6. Write $h(x)$ in vertex form.

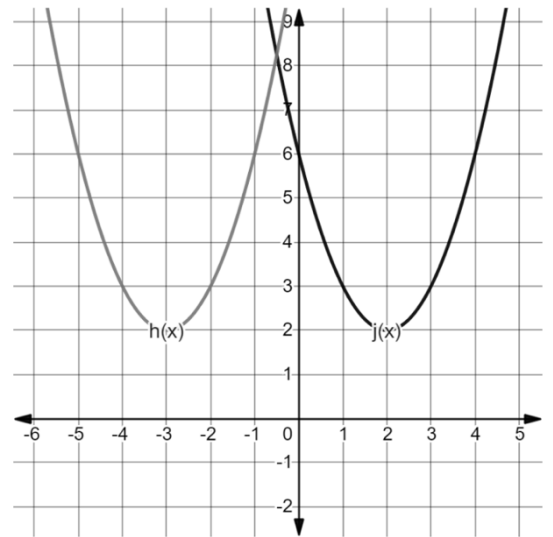
$$h(x) = (x + 3)^2 + 2$$

7. Write $j(x)$ in vertex form.

$$j(x) = (x - 2)^2 + 2$$

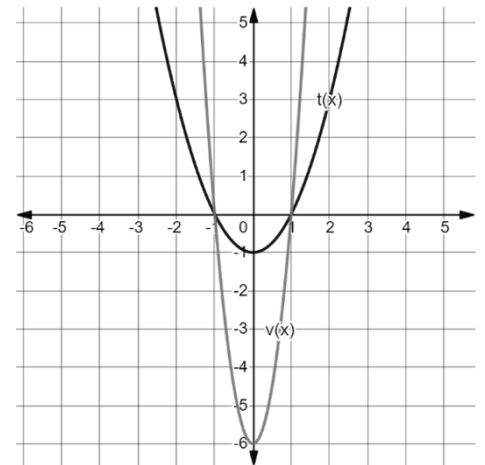
8. Describe how $h(x)$ was transformed to create $j(x)$.

$h(x)$ shifted 5 units right



9. The functions $t(x)$ and $v(x)$ are shown on the graph, where $v(x)$ is a transformation of $t(x)$. What is the value of k such that $v(x) = kt(x)$?

$$k = 6$$



10. The table of values for $w(x)$ and $z(x)$ are shown, where $z(x)$ is a transformation of $w(x)$. Write a description of the transformation.

$y = w(x)$		$y = z(x)$	
x	y	x	y
-2	8	-2	13
-1	2	-1	7
0	0	0	5
1	2	1	7
2	8	2	13

$w(x)$ shifted up 5 units which becomes $z(x)$.

Algebra 1

Unit 13

Lesson H2 Homework

Name Answer Key

Date _____

Period _____

Use the functions $k(x) = 3x^2 + 4x - 5$ and $p(x) = 6 - 2x$ to determine each new function. Write your answer in standard form.

1. $(k + p)(x)$

$$\begin{array}{r} 3x^2 + 4x - 5 \\ + \quad \quad - 2x + 6 \\ \hline 3x^2 + 2x + 1 \end{array} \quad (k + p)(x) = 3x^2 + 2x + 1$$

2. $(k - p)(x)$

$$\begin{array}{r} (3x^2 + 4x - 5) - (6 - 2x) \\ 3x^2 + 4x - 5 - 6 + 2x \end{array} \quad (k - p)(x) = 3x^2 + 6x - 11$$

3. $(p - k)(x)$

$$\begin{array}{r} (6 - 2x) - (3x^2 + 4x - 5) \\ 6 - 2x - 3x^2 - 4x + 5 \end{array} \quad (p - k)(x) = -3x^2 - 6x + 11$$

4. $(k \cdot p)(x)$

$$\begin{array}{r} (3x^2 + 4x - 5)(6 - 2x) \\ 18x^2 + 24x - 30 - 6x^3 - 8x^2 + 10x \end{array} \quad (k \cdot p)(x) = -6x^3 + 10x^2 + 34x - 30$$

5. $\left(\frac{p}{p}\right)x$

$$\frac{p}{p} = 1 \quad \left(\frac{p}{p}\right)x = 1$$

6. Write the restriction on the domain for $\left(\frac{p}{p}\right)x$.

$$6 - 2x = 0$$

$$-2x = -6$$

$$x = 3$$

$$x \neq 3$$

Juan and Marina are renting a car for a cross country road trip. The car costs \$155 per day. The rental includes 100 miles for free and then charges \$0.25 per mile. The function $C(m) = 0.25(m - 100) + 155$ models the cost of the car, C , in terms of miles traveled, m . They determine gas will cost \$1.15 for each mile they travel. The company charges a \$40 fee for the first tank of gas. The function $G(m) = 1.15m + 40$ models the cost of gasoline for their vacation.

7. What is a reasonable domain for $C(m)$?

Answers may vary.

$m \geq 0$ (for max: approx. miles from east to west coast 2,500 – 3,500 miles 1 way)

8. What is a reasonable domain for $G(m)$?

Answers may vary.

$m \geq 0$ (for max: $\frac{3,500 \cdot 2}{\text{avg mpg}}$)

9. Determine the function that gives the total cost, $T(m)$, in dollars, based on the miles traveled.

$$T(m) = C(m) + G(m)$$

$$0.25(m - 100) + 155 + (1.15m + 40) \quad \boxed{T(m) = 1.4m + 170}$$

$$0.25m - 25 + 155 + 1.15m + 40$$

$$1.4m + 170$$

10. What is a reasonable domain for $T(m)$?

Answers may vary.

$m \geq 0$